

Problem Sheet #1

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– Problem 1:

Problem 1 Show that

- i) The identity element of a group is unique.
- ii) The inverse of a given element of a group is unique.
- iii) $(gh)^{-1} = h^{-1}g^{-1}$ for any g, h elements of a group.

— *Solution:*

i) Let $e \in G$ be the identity element of G ,

thus we have the property $a * e = e * a = a$, $\forall a \in G$ Assume that there exist 2 elements with this properties s.t. $e_1 \neq e_2$.

Thus $e_1 * a = a \wedge e_2 * a = a$, by transitivity we have $e_1 * a = e_2 * a$, we could remove a from the equation, thus we have $e_1 = e_2$, which contradicts our assumption that there exist more than one element identity.

ii)